Stochastic Processes Semester 1

Tutorial 4 – Solutions

1. (a) Since $p_{02} > 0$, $p_{21} > 0$ and $p_{10} > 0$, we have $0 \to 2$, $2 \to 1$ and $1 \to 0$. It follows that all states intercommunicate since we can find a path from any state to any other state, and thus all states form a single closed class.

   (b) Since all states intercommunicate, by Theorem 4.4.4 there exists a unique stationary distribution. To find the stationary distribution, we solve the matrix equation $\boldsymbol{\pi} P = \boldsymbol{\pi}$, where $\boldsymbol{\pi} = (\pi_0, \pi_1, \pi_2)$ with the additional condition $\pi_0 + \pi_1 + \pi_2 = 1$. This gives

$$\begin{aligned}
\pi_0 &= \frac{1}{2}\pi_0 + \pi_1 + \frac{1}{4}\pi_2 \\
\pi_1 &= \frac{1}{2}\pi_2 \\
\pi_2 &= \frac{1}{2}\pi_0 + \frac{1}{4}\pi_2,
\end{aligned}$$

   and so the stationary distribution is given by $\boldsymbol{\pi} = (1/2, 1/6, 1/3)$.

   (c) For $n \geq 0$, let $\boldsymbol{\mu}^{(n)}$ be the distribution of X_n, i.e. $\mu_i^{(n)} = \mathbf{P}(X_n = i)$. We are given that

$$\boldsymbol{\mu}^{(0)} = (\mathbf{P}(X_n = 0), \mathbf{P}(X_n = 1), \mathbf{P}(X_n = 2)) = (1/3, 1/3, 1/3).$$

   Then $\boldsymbol{\mu}^{(n)} = \boldsymbol{\mu}^{(0)} P^n$ and so

$$\begin{aligned}
\boldsymbol{\mu}^{(1)} &= (7/12, 1/6, 1/4) \\
\boldsymbol{\mu}^{(2)} &= (25/48, 1/8, 17/48) \\
\boldsymbol{\mu}^{(3)} &= (91/192, 17/96, 67/192),
\end{aligned}$$

   so that the rate of convergence to the stationary distribution is fast.

2. (a) Recall that the required transition matrix is

$$P = \begin{pmatrix} 0 & 1/2 & 1/2 \\ 3/4 & 0 & 1/4 \\ 3/4 & 1/4 & 0 \end{pmatrix}.$$

   Solving $\boldsymbol{\pi} P = \boldsymbol{\pi}$ gives

$$\begin{aligned}
\pi_1 &= \frac{3}{4}\pi_2 + \frac{3}{4}\pi_3 \\
\pi_2 &= \frac{1}{2}\pi_1 + \frac{1}{4}\pi_3 \\
\pi_3 &= \frac{1}{2}\pi_1 + \frac{1}{4}\pi_2 \\
1 &= \pi_1 + \pi_2 + \pi_3,
\end{aligned}$$

   and so the stationary distribution of the taxi driver's location is

$$\boldsymbol{\pi} = (3/7, 2/7, 2/7).$$

   (b) The long term proportion of his journeys which terminate at a hotel is

$$\pi_2 + \pi_3 = 2/7 + 2/7 = 4/7.$$

3. Recall, here

$$P = \begin{pmatrix} 1/2 & 1/2 \\ 1 & 0 \end{pmatrix}.$$

Solving $\boldsymbol{\pi} P = \boldsymbol{\pi}$ along with $1 = \pi_1 + \pi_2$ gives

$$\pi_1 = \frac{1}{2}\pi_1 + \pi_2$$
$$\pi_2 = \frac{1}{2}\pi_1$$

so the stationary distribution $\boldsymbol{\pi}$ associated with the two-state Markov chain is $\boldsymbol{\pi} = (2/3, 1/3)$, and by Tutorial 2 Q3(b), we see that for all i, j, we have $\lim_{n\to\infty} p_{ij}^{(n)} = \pi_j$.

4. (a) The required transition matrix is

$$P = \begin{pmatrix} 0.15 & 0.85 & 0 \\ 0.15 & 0 & 0.85 \\ 0.01 & 0.14 & 0.85 \end{pmatrix}.$$

(b) The expected proportions are given by the stationary distribution

$$\boldsymbol{\pi} = (0.0352, 0.1447, 0.8201).$$

5. (a) Let the probabilities of 0, 1 and 2 or more claims in a year be q_0, q_1 and q_2, respectively. Then

$$\begin{aligned} q_0 &= \exp(-0.2) &&= 0.8187 \\ q_1 &= 0.2\exp(-0.2) &&= 0.1637 \\ q_2 &= 1 - q_0 - q_1 &&= 0.0176. \end{aligned}$$

The transition matrix of the Markov chain describing movement amongst the 4 levels, which we label $0, 1, 2, 3$ is

$$P = \begin{pmatrix} q_1+q_2 & q_0 & 0 & 0 \\ q_2 & q_1 & q_0 & 0 \\ q_2 & 0 & q_1 & q_0 \\ q_2 & 0 & 0 & q_1+q_0 \end{pmatrix} = \begin{pmatrix} 0.1813 & 0.8187 & 0 & 0 \\ 0.0176 & 0.1637 & 0.8187 & 0 \\ 0.0176 & 0 & 0.1637 & 0.8187 \\ 0.0176 & 0 & 0 & 0.9824 \end{pmatrix}.$$

[*Note that this is indeed a stochastic matrix.*]

(b) After one year the vector of the expected numbers at each level is

$$(10,000 \times p_{0j}^{(1)}) = (1813 \quad 8187 \quad 0 \quad 0).$$

After two years the vector of the expected numbers at each level is

$$(10,000 \times p_{0j}^{(2)}) = (472.8 \quad 2824.5 \quad 6702.7 \quad 0).$$

(c) Let $\boldsymbol{\pi}$ be the stationary distribution of the chain. Then the expected numbers at each discount level once stability has been achieved are given by

$$10,000\boldsymbol{\pi} = (210 \quad 206 \quad 202 \quad 9382).$$

6. (a) Each X_n is a random variable. The distribution of X_{n+1} depends on X_n , since shirt X_n is not available on day $n+1$, but not on previous X_js. In addition the state space is countable (indeed, it is finite). Hence $\{X_n\}$ is a Markov chain.

(b) Since there are 3 shirts to choose from for the next day, and each is equally likely, we have

$$\mathbf{P}(X_{n+1} = j \,|\, X_n = i) = \frac{1}{3}$$

for $i \neq j$, while $\mathbf{P}(X_{n+1} = i \,|\, X_n = i) = 0$. It follows that the transition matrix is

$$P = \begin{pmatrix} 0 & 1/3 & 1/3 & 1/3 \\ 1/3 & 0 & 1/3 & 1/3 \\ 1/3 & 1/3 & 0 & 1/3 \\ 1/3 & 1/3 & 1/3 & 0 \end{pmatrix}$$

from which the required result

$$P = \frac{4}{3}E - \frac{1}{3}I$$

is immediate.

(c) The given formula has already been established in the case $n = 1$. Suppose now that it is true for $n = k$. Then

$$\begin{aligned} P^{k+1} &= P^k P \\ &= \left(1 - \left(-\frac{1}{3}\right)^k\right) EP + \left(-\frac{1}{3}\right)^k IP \\ &= \left(1 - \left(-\frac{1}{3}\right)^k\right) \left(\frac{4}{3}E^2 - \frac{1}{3}E\right) + \left(-\frac{1}{3}\right)^k \left(\frac{4}{3}E - \frac{1}{3}I\right) \\ &= \left(1 - \left(-\frac{1}{3}\right)^k\right) E + \left(-\frac{1}{3}\right)^k \left(\frac{4}{3}E - \frac{1}{3}I\right) \\ &= \left(1 - \left(-\frac{1}{3}\right)^{k+1}\right) E + \left(-\frac{1}{3}\right)^{k+1} I \end{aligned}$$

since $E^2 = E$. Hence the required induction is established.

(d) It is clear by symmetry that the stationary distribution is given by $\boldsymbol{\pi} = (1/4, 1/4, 1/4, 1/4)$. This may be verified, if you like, by checking that this $\boldsymbol{\pi}$ satisfies the stationary equations, and then appealing to the uniqueness of the solution to the latter. Further $\lim_{n\to\infty} P^n = E$, as required.

(e) $f_{ii}(1) = 0$ since he wears a different shirt each day. For $n > 1$, with probability 1 he does not wear the shirt again the following day, and then on any particular day he does not wear this shirt with probability $2/3$ and if the lecturer has not worn the shirt i on day $n-1$, there is a probability of $1/3$ that he will wear it on day n.

We now have that mean number of days between successive wearings of the shirt i is

$$\begin{aligned} \sum_{n\geq 1} n f_{ii}(n) &= \sum_{n\geq 2} n \left(\frac{2}{3}\right)^{n-2} \frac{1}{3} = \frac{3}{2} \sum_{n\geq 2} n \left(\frac{2}{3}\right)^{n-1} \frac{1}{3} \\ &= \frac{3}{2} \left(\sum_{n\geq 1} n \left(\frac{2}{3}\right)^{n-1} \frac{1}{3} \right) - \frac{1}{2} = \frac{9}{2} - \frac{1}{2} = 4. \end{aligned}$$

7. Let the transition matrix be $P = (p_{ij})$. Then, since it is doubly stochastic,

$$\sum_{i=1}^{N} p_{ij} = 1 = \sum_{j=1}^{N} p_{ij}.$$

Thus

$$\begin{aligned}\frac{1}{N}(1,1,..,1)P &= \frac{1}{N}\left(\sum_{i=1}^{N} p_{i1}, \sum_{i=1}^{N} p_{i2}, \ldots, \sum_{i=1}^{N} p_{iN}\right) \\ &= \frac{1}{N}(1,1,...,1),\end{aligned}$$

and so $N^{-1}(1,...,1)$ is a stationary distribution for the chain.

8. (a) The possible values for X_n are $0, 1, ..., N$. Suppose $X_n = i$, so that also after the nth swap there are $N - i$ black balls in the second urn. Then, for $i = 1, 2, 3, .., N - 1$,

$$X_{n+1} = \begin{cases} i & \text{if either two white or two black balls are selected} \\ i+1 & \text{if a white ball is selected from urn 1 and a black ball is selected from urn 2} \\ i-1 & \text{if a black ball is selected from urn 1 and a white ball is selected from urn 2.} \end{cases}$$

Hence the transition matrix P of the chain is given as follows.
For $i = 1, 2, 3, .., N - 1$,

$$\begin{aligned} p_{ii} &= \frac{N-i}{N} \times \frac{i}{N} + \frac{i}{N} \times \frac{N-i}{N} = \frac{2i(N-i)}{N^2} \\ p_{i,i+1} &= \frac{N-i}{N} \times \frac{N-i}{N} = \left(\frac{N-i}{N}\right)^2 \\ p_{i,i-1} &= \frac{i}{N} \times \frac{i}{N} = \left(\frac{i}{N}\right)^2. \end{aligned}$$

For $i = 0$ we have

$$p_{00} = 0; \qquad p_{01} = 1,$$

since a white ball must be selected from urn 1 and a black ball must be selected from urn 2.
For $i = N$ we similarly have

$$p_{NN} = 0; \qquad p_{N,N-1} = 1.$$

All other entries of P will be 0 since we can only reduce or increase the number of black balls in urn by at most 1 ball in a single swap.

(b) Note that with enough swaps, it is possible to go from 0 balls in urn 1 up to N balls in urn 1, and vice versa. Hence all states intercommunicate and, since the state space is finite (of size $N + 1$), there will exist a unique stationary distribution.

We first show that the given vector $\boldsymbol{\pi}$ satisfies $\boldsymbol{\pi}P = \boldsymbol{\pi}$. Note that $\boldsymbol{\pi} = k\boldsymbol{\mu}$ where

$$\mu_j = \frac{1}{(j!)^2((N-j)!)^2}$$

and the constant k is independent of j.

Hence it is sufficient (and easier) to show that $\boldsymbol{\mu}P = \boldsymbol{\mu}$, i.e.

$$\sum_{i \in S} \mu_i p_{ij} = \mu_j$$

(where $S = \{0, \ldots, N\}$ and P is as given above). So, for $0 < j < N$,

$$\begin{aligned}
\sum_{i \in S} \mu_i p_{ij} &= \mu_{j-1} p_{j-1\,j} + \mu_j p_{jj} + \mu_{j+1} p_{j+1\,j} \\
&= \left(\frac{1}{((j-1)!)^2((N-j+1)!)^2} \right) \left(\frac{N-j+1}{N} \right)^2 \\
&+ \left(\frac{1}{(j!)^2((N-j)!)^2} \right) \left(\frac{2j(N-j)}{N^2} \right) \\
&+ \left(\frac{1}{((j+1)!)^2((N-j-1)!)^2} \right) \left(\frac{j+1}{N} \right)^2 \\
&= \frac{1}{(j!)^2((N-j)!)^2} \\
&= \mu_j
\end{aligned}$$

We now need to check that $\sum_{j=0}^{N} \pi_j = 1$. This can be verified by considering the coefficient of x^N in the binomial expansions

$$(1+x)^N (1+x)^N = \left(\sum_{i=0}^{N} \binom{N}{i} x^i \right) \left(\sum_{j=0}^{N} \binom{N}{j} x^j \right)$$

and

$$(1+x)^{2N} = \left(\sum_{j=0}^{2N} \binom{2N}{j} x^j \right),$$

since these two expressions will be equal, and recalling that $\binom{N}{j} = \binom{N}{N-j}$.

9. For $\boldsymbol{\pi}$ as given, first we check $\sum_{i=0}^{\infty} \pi_i = 1$. So,

$$\sum_{i=0}^{\infty} \pi_i = \sum_{i=0}^{\infty} \left(1 - \frac{p}{q}\right) \left(\frac{p}{q}\right)^i = \left(1 - \frac{p}{q}\right) \sum_{i=0}^{\infty} \left(\frac{p}{q}\right)^i = \left(1 - \frac{p}{q}\right) \left(\frac{1}{1 - p/q}\right) = 1.$$

We must also show that $\boldsymbol{\pi}$ satisfies the following stationarity equations:

$$\begin{aligned} \pi_0 q + \pi_1 q &= \pi_0 \\ \pi_{i-1} p + \pi_{i+1} q &= \pi_i \qquad\qquad i = 1, 2, 3, \ldots . \end{aligned}$$

So,

$$\begin{aligned} \pi_0 q + \pi_1 q &= \left(1 - \frac{p}{q}\right) q + \left(1 - \frac{p}{q}\right) \left(\frac{p}{q}\right) q = \frac{q^2 - p^2}{q} \\ &= \frac{(q-p)(q+p)}{q} = \frac{q-p}{q} = 1 - \frac{p}{q} = \pi_0 \end{aligned}$$

and for $i > 0$

$$\begin{aligned} \pi_{i-1} p + \pi_{i+1} q &= \left(1 - \frac{p}{q}\right) \left(\frac{p}{q}\right)^{i-1} p + \left(1 - \frac{p}{q}\right) \left(\frac{p}{q}\right)^{i+1} q \\ &= \left(1 - \frac{p}{q}\right) \left(\frac{p}{q}\right)^{i-1} \left(p + \frac{p^2}{q}\right) \\ &= \left(1 - \frac{p}{q}\right) \left(\frac{p}{q}\right)^{i-1} \left(\frac{pq + p^2}{q}\right) \\ &= \left(1 - \frac{p}{q}\right) \left(\frac{p}{q}\right)^{i-1} \left(\frac{p((1-p) + p)}{q}\right) \\ &= \left(1 - \frac{p}{q}\right) \left(\frac{p}{q}\right)^{i} = \pi_i. \end{aligned}$$